

Countering Simpson's Paradox with Counterfactuals

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Introduction

- Aggregation is a powerful tool to show summary statistics in visualizations.
- However, it can also introduce additional risks, such as Simpson's Paradox — trends that appear at one level of aggregation may disappear or reverse when data is subdivided into lower levels of aggregation.

Simpson's Paradox

Stone Size	T_A	T_B
Small	93% (81/87)	87% (234/270)
Large	73% (192/263)	69% (55/80)
All	78% (273/350)	83% (289/350)

The above Kidney Stone study included patients with stones of variable size, classified as large or small.

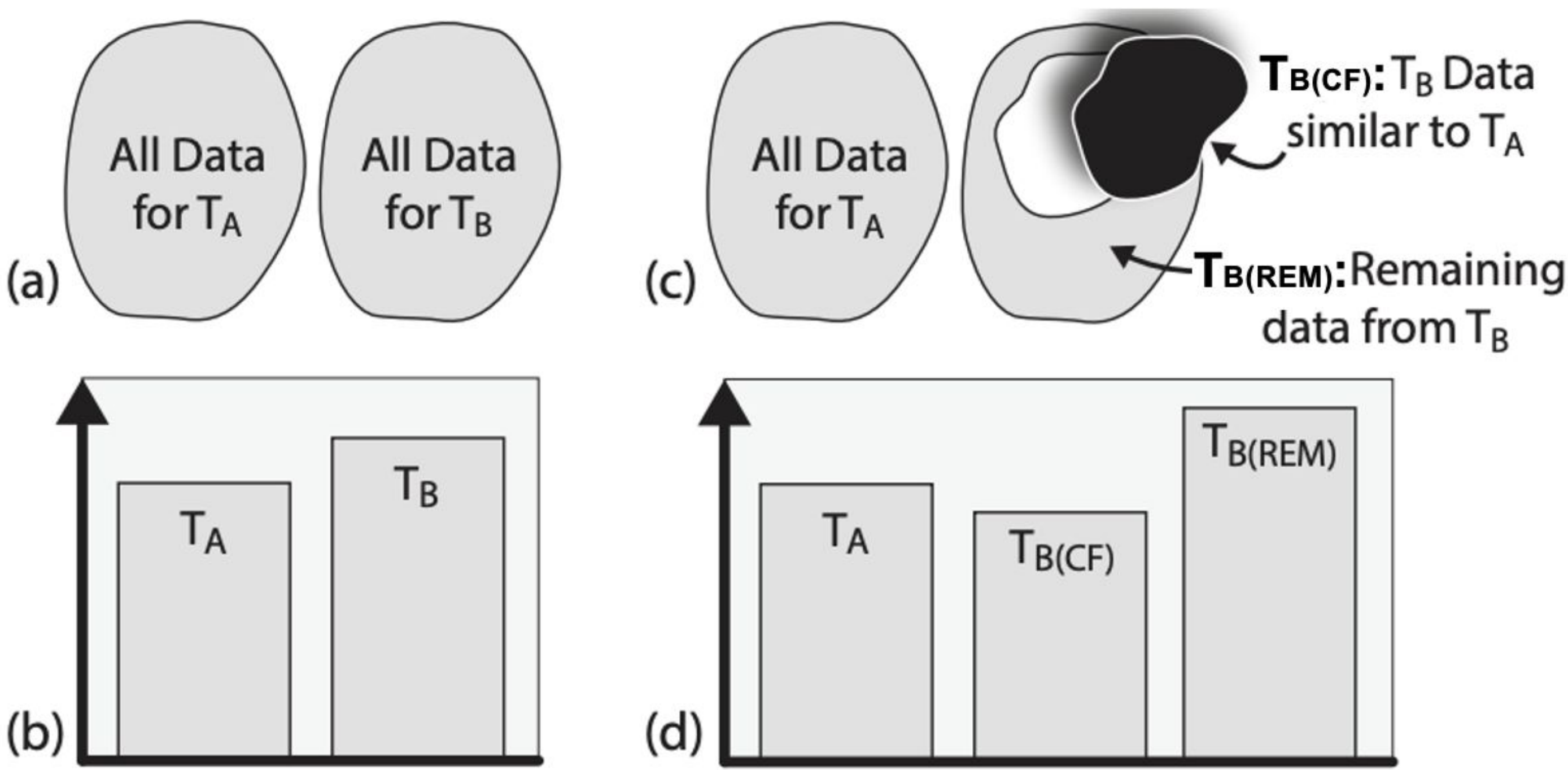
Compared to **Treatment B (T_B)**, **Treatment A (T_A)** performed best on small stones and best on large stones. However, counter-intuitively, **Treatment B** appeared to have a higher success rate overall, as a result of the *unequal distribution of patient groups*.

Visualization of Counterfactuals

Counterfactual reasoning, by constructing **hypothetical scenarios** ("what if things were the same except for this one fact?"), can be used to overcome the imbalance between the distributions.

Counterfactuals can be simulated by sampling from the population receiving **T_B** a subset of patients similar to those patients receiving **T_A** , referred to as **$T_{B(CF)}$** .

$T_{B(CF)}$ will comprise a group of patients with similar variable distributions to **T_A** . In the Kidney example, we sample a group of patients from **T_B** with the same ratio of *large:small* kidney stones as **T_A** to include in **$T_{B(CF)}$** . The remainder samples are noted as **$T_{B(REM)}$** .



(a-b) includes a typical visualization comparing **T_A** and **T_B** without considering counterfactuals.

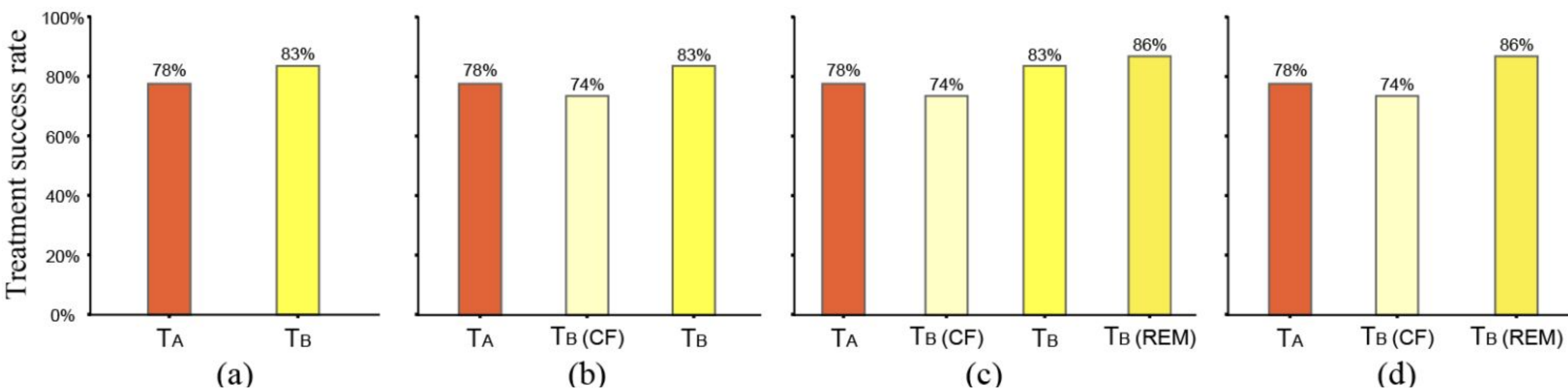
(c) shows the construction of counterfactual subset **$T_{B(CF)}$** and remainder subset **$T_{B(REM)}$** .

(d) shows a visualization comparing **T_A** and **$T_{B(CF)}$** which can avoid Simpson's Paradox.

Countering Simpson's Paradox

Stone Size	T_A	T_B	$T_{B(CF)}$	$T_{B(REM)}$
Small	93% (81/87)	87% (234/270)	88% (23/26)	86% (211/244)
Large	73% (192/263)	69% (55/80)	69% (55/80)	N/A (0/0)
All	78% (273/350)	83% (289/350)	74% (78/106)	86% (211/244)

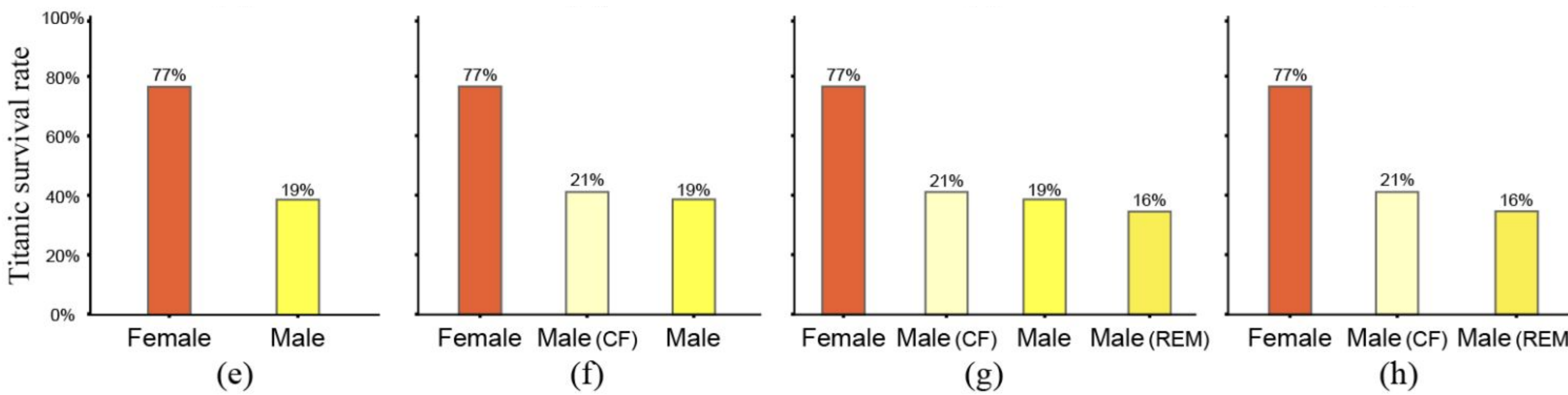
Constructed subsets of Kidney Stone dataset.
w/ Simpson's Paradox



Compared to a traditional visualization (a) of the two treatments, designs that incorporate counterfactuals (b-d) can more accurately communicate the desired comparison (**$T_{B(CF)}$** is worse than **T_A**) between treatments.

Cabin	Female	Male	$Male_{(CF)}$	$Male_{(REM)}$
Class 1	97% (91/94)	37% (45/122)	37% (35/94)	36% (10/28)
Class 2	92% (70/76)	16% (17/108)	16% (12/76)	16% (5/32)
Class 3	56% (81/144)	14% (47/347)	14% (20/144)	13% (27/203)
All	77% (242/314)	19% (109/577)	21% (67/314)	16% (42/263)

Constructed subsets of Titanic Survival dataset.
w/o Simpson's Paradox



When Simpson's Paradox is not present, a traditional visualization (e) and designs that incorporate counterfactuals (f-h) can both accurately communicate the desired comparison (**$Male_{(CF)}$** is lower than **$Female$**).